

FlowServe XR-900 Centrifugal Pump

Operation, Maintenance and Diagnostic Manual | Rev 4.2
Tag Reference: P-101 | Functional Area: Cooling Infrastructure

1. Product Overview

The FlowServe XR-900 is a heavy-duty centrifugal pump for continuous industrial cooling. It is engineered for high-temperature fluid transfer in manufacturing and process environments. The XR-900 features a dual-mechanical seal, corrosion-resistant impeller housing, and integrated vibration sensor mounts compatible with ISO 10816-3 standard monitoring. Model: XR-900 | Manufacturer: FlowServe | Asset Tag: P-101

2. Technical Specifications

Parameter	Specification	Unit
Rated Flow Rate	450	m3/hr
Max Operating Pressure	12.5	bar
Rated Speed	2950	RPM
Motor Power	75	kW
Impeller Diameter	320	mm
Shaft Torque (rated)	243	N.m
Shaft Torque (max)	310	N.m
Operating Temp Range	-10 to 180	degC
NPSH Required	4.2	m
Bearing Type	SKF 6316	-
Seal Type	Dual mechanical Type-B	-
Base Emissions CO2	45	ppm

3. Installation Requirements

Mounting: Install on level concrete base with anchor bolts torqued to 80 N.m.
Alignment: Max shaft misalignment 0.05 mm radial, 0.02 mm angular.
Upstream Valve: Isolation valve V-20 (Swagelok V-LOK-2) must be installed on suction line. V-20 controls flow into P-101 and is critical for priming and cavitation prevention. IMPORTANT: V-20 must be fully open during normal pump operation. A closed or partially closed V-20 will restrict inlet flow and cause cavitation in P-101.
Discharge: Downstream feeds into Boiler B-04 heat exchanger feed line. P-101 is the sole feed pump for B-04. If P-101 goes offline, B-04 must shut down within 8 minutes to prevent thermal runaway.

XR-900 Maintenance and Diagnostic Guide

4. Scheduled Maintenance Intervals

Interval	Task	Ref
250 hrs	Lubricate bearing housing with Shell Gadus	PM-XR9-001
500 hrs	Inspect mechanical seal for leakage; replace	PM-XR9-002
1000 hrs	Full impeller inspection; measure wear ring	PM-XR9-003
2000 hrs	Replace shaft bearing set (SKF 6316)	PM-XR9-004
5000 hrs	Full pump overhaul: impeller, seals, bearings	PM-XR9-005

5. Vibration Diagnostic Guide

Baseline vibration at rated load: 1.8 mm/s RMS.
Alarm threshold: 4.5 mm/s. Trip threshold: 7.1 mm/s.

SYMPTOM: Elevated vibration (3 to 6 mm/s) during high thermal load periods.
ROOT CAUSE: Partial flow starvation because upstream valve V-20 is partially closed.
Under high thermal load the fluid density drops, increasing required NPSH.
If V-20 is not fully open, available NPSH falls below required 4.2 m causing cavitation.
Cavitation produces high-frequency vibration in the 1 to 10 kHz band.

- CORRECTIVE ACTION:
1. Check upstream valve V-20 (Swagelok V-LOK-2 on suction line).
 2. If V-20 is less than fully open, open it to 100% via handwheel.
Standard correction: open V-20 by 1.5 to 2.0 full rotations counterclockwise.
 3. Monitor vibration for 5 minutes. Should return below 2.0 mm/s.
 4. If vibration persists after V-20 check, inspect seal per PM-XR9-002.

SYMPTOM: Flow rate drop below 300 m3/hr.
ROOT CAUSE A: V-20 closed or restricted. Check and correct immediately.
ROOT CAUSE B: Impeller wear. Schedule PM-XR9-003 if V-20 is fully open.

SYMPTOM: Overheating, bearing housing temperature above 85 degC.
CAUSE: Lubrication failure or misalignment causing excessive radial load.
ACTION: Lubricate per PM-XR9-001. Verify coupling alignment.

6. Emergency Shutdown Procedure

- Initiate emergency shutdown if:
- Vibration exceeds 7.1 mm/s trip threshold
 - Bearing temperature exceeds 95 degC
 - Mechanical seal leak detected
 - Discharge pressure drops more than 40% below rated for more than 30 seconds

- SHUTDOWN STEPS:
1. Close discharge valve before stopping motor.
 2. De-energize motor via local isolator.
 3. Close upstream valve V-20 to isolate pump from suction line.
 4. Notify B-04 boiler operator: P-101 shutdown removes coolant feed to B-04.
B-04 must initiate controlled shutdown within 8 minutes.
 5. Log incident in CMMS with vibration readings and timestamp.

XR-900 Torque Reference and Process Integration

7. Bolt Torque Specifications

Location	Bolt Size	Torque (N.m)	Pattern
Casing split bolts	M20	180	Cross pattern
Bearing housing	M16	110	Star pattern
Impeller lock nut	M30	420	Single
Mechanical seal gland	M12	45	Even cross
Base plate anchor	M24	310	All corners first
Coupling hub bolts	M16	130	Alternating
Suction flange (V-20)	M20	165	Cross pattern
Discharge flange	M20	165	Cross pattern

8. Process Integration Notes

P-101 is part of a three-component cooling loop:

[V-20 Swagelok V-LOK-2] --> [P-101 FlowServe XR-900] --> [B-04 Thermax BLR-MAX]
upstream downstream

V-20 is the primary flow control and isolation point for P-101 suction line.
P-101 is the sole coolant feed pump for the Thermax BLR-MAX boiler (B-04).
B-04 generates 320 ppm CO2 equivalent under full load.

OPERATIONAL DEPENDENCY CHAIN:
If V-20 is closed or restricted: P-101 cavitates or loses flow.
If P-101 trips: B-04 loses coolant feed and must shut down within 8 minutes.
If B-04 shuts down: Power-Gen functional area loses primary steam source.

All maintenance on any component in this chain requires coordination across Cooling and Power-Gen areas. A permit-to-work must be raised for both areas.

9. Spare Parts Minimum Stock

Part Number	Description	Min Qty
XR9-IMP-320	Impeller assembly, 320mm, S316 steel	1
XR9-SEAL-B	Dual mechanical seal kit, Type-B	2
XR9-BRG-6316	Shaft bearing set, SKF 6316	2
XR9-WR-CLR	Wear ring set (suction + discharge)	2
XR9-CPL-FLX	Flexible coupling element, 75kW rated	1